



**The CENTRE for EDUCATION
in MATHEMATICS and COMPUTING**

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**2025 Gauss Grade 7
Solutions**

1. In a group of 12 friends, each friend gives \$5 to a charity. How much money does the group give in total?

(A) \$50 (B) \$55 (C) \$60 (D) \$65 (E) \$70

Source: 2025 Gauss Grade 7 #1

Primary Topics: Number Sense

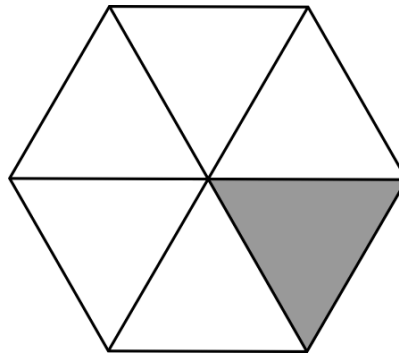
Secondary Topics: Operations

Answer: C

Solution:

If each of 12 friends gives \$5, in total they give $12 \times \$5 = \60 to the charity.

2. In the diagram, what fraction of the area of the regular hexagon is shaded?



(A) $\frac{1}{3}$ (B) $\frac{1}{4}$ (C) $\frac{1}{7}$ (D) $\frac{1}{2}$ (E) $\frac{1}{6}$

Source: 2025 Gauss Grade 7 #2

Primary Topics: Geometry and Measurement | Number Sense

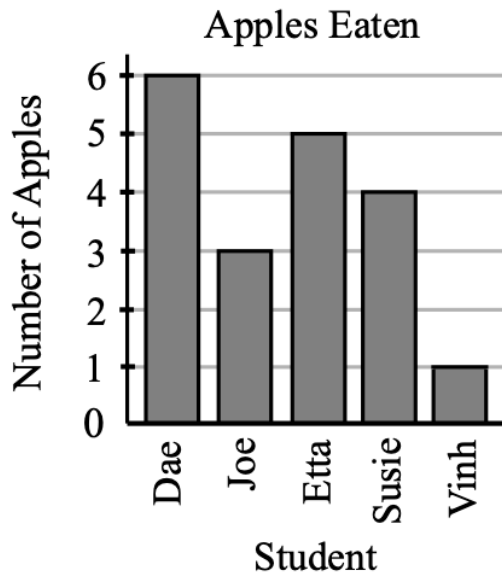
Secondary Topics: Polygons | Fractions/Ratios

Answer: E

Solution:

The regular hexagon is divided into 6 triangles having equal area. Since 1 of these 6 triangles is shaded, then $\frac{1}{6}$ of the area of the hexagon is shaded.

3. The graph shows the number of apples that each of five students ate during a week.



Which student ate the greatest number of apples?

- (A) Dae (B) Joe (C) Etta (D) Susie (E) Vinh

Source: 2025 Gauss Grade 7 #3

Primary Topics: Data Analysis

Secondary Topics: Counting | Area

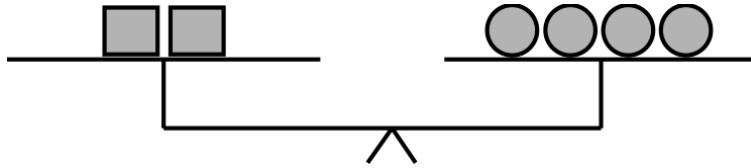
Answer: A

Solution:

Reading from the graph, Dae ate 6 apples, Joe ate 3, Etta ate 5, Susie ate 4, and Vinh ate 1. Thus, Dae ate the greatest number of apples.

(We may have simply noted that the person who ate the greatest number of apples is the person with the 'highest bar'.)

4. The equal-arm scale shown is balanced.



One  has the same mass as

- (A) 
- (B) 
- (C) 
- (D) 
- (E) 

Source: 2025 Gauss Grade 7 #4

Primary Topics: Data Analysis

Secondary Topics: Counting | Equations Solving

Answer: B

Solution:

The equal-arm scale shows that 2 squares have the same mass as 4 circles.

If we split the 2 squares and the 4 circles each into two equal groups, then 1 square has the same mass as 2 circles.

5. Which of the following is equal to the area of a square with side length 8?

- (A) 8×2
- (B) $2 \times (8 + 8)$
- (C) 8×8
- (D) 4×8
- (E) $8 \times 8 \times 8 \times 8$

Source: 2025 Gauss Grade 7 #5

Primary Topics: Geometry and Measurement

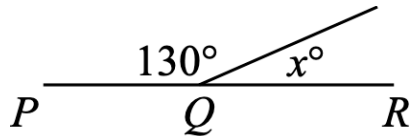
Secondary Topics: Measurement | Area | Quadrilaterals | Polygons

Answer: C

Solution:

A square with side length 8 has area $8 \times 8 = 64$. Since $8 \times 2 = 16$, $2 \times (8 + 8) = 2 \times 16 = 32$, $4 \times 8 = 32$, and $8 \times 8 \times 8 \times 8 = 4096$, then 8×8 is the only expression given that is equal to the area of the square.

6. In the diagram, $\angle PQR$ is a straight angle.



The value of x is

- (A) 60 (B) 40 (C) 50 (D) 70 (E) 65

Source: 2025 Gauss Grade 7 #6

Primary Topics: Geometry and Measurement

Secondary Topics: Angles

Answer: C

Solution:

Since $\angle PQR$ is a straight angle, its measure is 180° . Thus, the angles with measures 130° and x° add to 180° , or $130 + x = 180$ and so $x = 180 - 130 = 50$.

7. The list of seven numbers 3, 15, 8, 8, 9, 9, n has exactly one mode, which is 8. What is the value of n ?

- (A) 15 (B) 9 (C) 3 (D) 8 (E) 10

Source: 2025 Gauss Grade 7 #7

Primary Topics: Data Analysis | Counting and Probability

Secondary Topics: Averages | Counting

Answer: D

Solution:

In a list of numbers having exactly one mode, the mode is the number that appears in the list the greatest number of times. Since the mode of the given list is 8, and the list has exactly one mode, then 8 must appear more times than each of the other numbers in the list.

If $n = 8$, then 8 appears in the list three times, which is more than any other number in the list, and so the list has exactly one mode, which is 8, as required.

If the value of n is 15 or 3, then the list has three numbers that each appear twice, and thus the list does not have exactly one mode.

If $n = 9$, then 9 appears more times than any other number and so the mode is 9.

If $n = 10$, then both 8 and 9 each appear twice (which is more than any other number in the list), and thus the list does not have exactly one mode.

8. Sam has only one measuring container. The volume of this container is $\frac{1}{2}$ cup. A recipe needs $2\frac{1}{2}$ cups of flour. How many times does Sam fill his $\frac{1}{2}$ cup container to accurately measure the flour for this recipe?
- (A) 3 (B) $2\frac{1}{2}$ (C) 10 (D) $4\frac{1}{2}$ (E) 5

Source: 2025 Gauss Grade 7 #8

Primary Topics: Number Sense

Secondary Topics: Counting | Fractions/Ratios

Answer: E

Solution:

Solution 1:

To measure 1 cup of flour, Sam fills his $\frac{1}{2}$ cup container 2 times (since $\frac{1}{2} \times 2 = 1$).

To measure 2 cups of flour, Sam fills his $\frac{1}{2}$ cup container $2 \times 2 = 4$ times.

To measure $2\frac{1}{2}$ cups of flour, Sam fills his $\frac{1}{2}$ cup container $4 + 1 = 5$ times.

Solution 2:

Since $2\frac{1}{2}$ cups is equal to $\frac{5}{2}$ cups, and $5 \times \frac{1}{2} = \frac{5}{2}$, then Sam fills his $\frac{1}{2}$ cup container 5 times.

9. The month of June has 30 days. If in a certain year June 1 is on a Tuesday, on which day of the week is June 30?
- (A) Monday (B) Tuesday (C) Wednesday (D) Thursday (E) Friday

Source: 2025 Gauss Grade 7 #9

Primary Topics: Number Sense | Counting and Probability

Secondary Topics: Counting | Patterning/Sequences/Series

Answer: C

Solution:

There are 7 days in a week. If June 1 is a Tuesday, then by moving successively 7 days later in the month, we get June 8, 15, 22, and 29 are also Tuesdays. Thus, June 30 is a Wednesday.

10. The words "P U G F O R S A L E" are written on a store window. How many of these ten letters look the same when viewed from both sides of the window?
- (A) 3 (B) 4 (C) 5 (D) 6 (E) 7

Source: 2025 Gauss Grade 7 #10

Primary Topics: Geometry and Measurement

Secondary Topics: Transformations

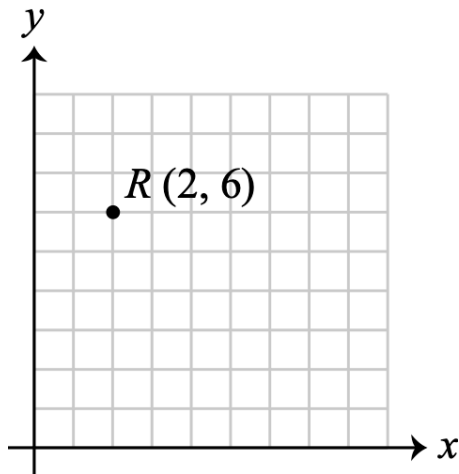
Answer: A

Solution:

When viewed from the opposite side of the window, text appears reflected horizontally.

That is, P U †G †F †O R S A L E appears as . The letters that look the same when viewed from both sides of the window are A, O and U and so there are 3 such letters.

11. The coordinates of R are $(2, 6)$, as shown. After which of these translations will R move to the point $(7, 0)$?



- (A) right (9), down (6) (B) left (5), up (6) (C) right (6), down (5)
(D) left (6), down (5) (E) right (5), down (6)

Source: 2025 Gauss Grade 7 #11

Primary Topics: Geometry and Measurement

Secondary Topics: Coordinate Geometry | Transformations

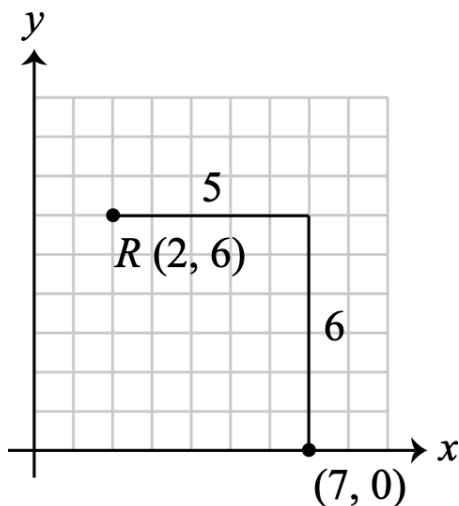
Answer: E

Solution:

The x -coordinate of $(2, 6)$ is 5 less than the x -coordinate of $(7, 0)$, and so R must be translated right 5.

The y -coordinate of $(2, 6)$ is 6 more than the y -coordinate of $(7, 0)$, and so R must be translated down 6.

Thus, the translation right 5, down 6 will move $R(2, 6)$ to $(7, 0)$.



12. A train stops at Waterloo Station every 3 minutes. A bus stops at Waterloo Station every 5 minutes. A train and a bus both stop at Waterloo Station at 6 : 25 *a. m.* The next time that a train and a bus both stop at Waterloo Station at the same time is
(A) 6 : 28 *a. m.* (B) 6 : 30 *a. m.* (C) 6 : 33 *a. m.* (D) 6 : 40 *a. m.* (E) 6 : 55 *a. m.*

Source: 2025 Gauss Grade 7 #12

Primary Topics: Number Sense | Counting and Probability

Secondary Topics: Counting | Patterning/Sequences/Series

Answer: D

Solution:

A train stops at Waterloo Station at 6:25 a.m. and every 3 minutes thereafter, and so a train stops only at times which differ from 6:25 a.m. by a multiple of 3. Of the given answers, a train stops at the station at 6:28 a.m. (difference is 3 minutes), at 6:40 a.m. (difference is 15 minutes), and at 6:55 a.m. (difference is 30 minutes).

A bus stops at Waterloo Station at 6:25 a.m. and every 5 minutes thereafter, and so a bus stops only at times which differ from 6:25 a.m. by a multiple of 5. Of the given answers, a bus stops at the station at 6:30 a.m. (difference is 5 minutes), at 6:40 a.m. (difference is 15 minutes), and at 6:55 a.m. (difference is 30 minutes).

Thus, the next time that both a train and a bus stop at Waterloo Station at the same time is 6:40 a.m.

Note: The lowest common multiple of both 3 and 5 is 15. This tells us that a train and a bus will stop at the station every 15 minutes after 6:25 a.m. This confirms that 6:40 a.m. is the next time that both a train and a bus will stop at the station at the same time.

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13. The numbers 2, 0, 2, 5 are repeated to form the pattern 2, 0, 2, 5, 2, 0, 2, 5, ...
If a total of 50 numbers are written, how many times will the number 5 appear?
(A) 10 (B) 11 (C) 12 (D) 13 (E) 25

Source: 2025 Gauss Grade 7 #13

Primary Topics: Counting and Probability

Secondary Topics: Patterning/Sequences/Series | Counting

Answer: C

Solution:

The pattern repeats in blocks of the 4 numbers 2, 0, 2, 5. Since $50 = 12 \times 4 + 2$, then the 50 numbers in the pattern contain 12 complete blocks of 2, 0, 2, 5 followed by the first 2 numbers in the block, 2, 0.

The number 5 appears once in each of the 12 blocks and does not appear as the 49th or 50th number. Thus, the number 5 appears 12 times.

14. What number goes in the box so that $\frac{28}{32} + \frac{1}{\square} = 1$?
- (A) 24 (B) -3 (C) 7 (D) 16 (E) 8

Source: 2025 Gauss Grade 7 #14

Primary Topics: Algebra and Equations

Secondary Topics: Equations Solving | Fractions/Ratios

Answer: E

Solution:

Since $\frac{28}{32} + \frac{1}{\square} = 1$ and $\frac{28}{32} + \frac{4}{32} = \frac{32}{32} = 1$, then $\frac{1}{\square} = \frac{4}{32}$.

Reducing $\frac{4}{32}$ to lowest terms, we get $\frac{4}{32} = \frac{1}{8}$, and so the number that goes in the box is 8.

15. Two standard six-sided dice are rolled. If the two numbers on the top faces are added, which of the following sums is least likely?
- (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

Source: 2025 Gauss Grade 7 #15

Primary Topics: Counting and Probability

Secondary Topics: Probability | Games

Answer: E

Solution:

In the grid shown, the first column and first row show the possible numbers that may appear on the top faces of the two dice. The inside of the grid shows the sum of those corresponding two numbers.

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Of the 36 possible combinations of rolls, a sum of 7 can happen 6 different ways, a sum of 8 can happen 5 different ways, a sum of 9 can happen 4 different ways, a sum of 10 can happen 3 different ways, and a sum of 11 can happen 2 different ways. Thus, of the given sums, 11 is the least likely to occur.

16. Each of the digits 7, 1, 3, 6, 8, and 2 is placed into one of the squares below to make an expression containing three 2-digit numbers.

$$\square\square + \square\square - \square\square$$

When the first two 2-digit numbers are added and the third is subtracted, the greatest possible result is

- (A) 139 (B) 145 (C) 147 (D) 149 (E) 138

Source: 2025 Gauss Grade 7 #16

Primary Topics: Number Sense

Secondary Topics: Operations | Optimization | Digits

Answer: C

Solution:

For the greatest possible result, we want the sum of the first two 2-digit numbers to be as large as possible, and the third 2-digit number to be as small as possible.

Since the tens digit contributes more to the value of a number than the ones digit, we choose the two largest digits, 7 and 8, as the tens digits of the first two 2-digit numbers, and the smallest of the given digits, 1, as the tens digit of the third 2-digit number.

Of the remaining digits, 3, 6, and 2, we choose the two largest digits, 3 and 6, as the ones digits of the first two 2-digit numbers, and 2 as the ones digit of the third 2-digit number, which becomes 12.

Since $73 + 86 = 76 + 83$, it does not matter how we pair the tens digits with the ones digits for the first two 2-digit numbers.

Thus, the greatest possible result is $73 + 86 - 12 = 147$.

17. Savannah tossed a fair coin some number of times and 50% of those tosses resulted in tails. She then tossed the coin one final time and the result was tails. If 60% of all tosses resulted in tails, how many tosses did she make in total?
 (A) 3 (B) 9 (C) 8 (D) 5 (E) 10

Source: 2025 Gauss Grade 7 #17

Primary Topics: Counting and Probability

Secondary Topics: Counting | Probability | Games

Answer: D

Solution:

Solution 1:

We begin by recognizing that written as a fraction in lowest terms, 60% is equal to $\frac{3}{5}$.

Thus, if Savannah tossed 3 tails out of 5 tosses, then 60% of the tosses that she made were tails.

Since the result of her last toss was tails, then following Savannah's second last toss she had tossed 2 tails out of 4 tosses, meaning that 50% of the tosses that she made to that point were tails, as required.

Therefore, Savannah made 5 tosses in total.

Solution 2:

We proceed to work backward from the answers given.

If Savannah made 3 tosses, and 60% of those tosses were tails, then she tossed $3 \times \frac{60}{100} = \frac{180}{100} = 1.8$ tails. She must toss a whole number of tails, and so this is not possible.

If Savannah made 9 tosses, then she tossed $9 \times \frac{60}{100} = \frac{540}{100} = 5.4$ tails. Again, this is not possible.

If Savannah made 8 tosses, then her second last toss was her 7th toss. However, it is not possible for Savannah to toss 50% tails in 7 tosses (since one half of 7 is not a whole number). (We could have instead shown that 60% of 8 is also not an integer as we did in the first two cases.)

Similarly, Savannah could not have made 10 tosses since one half of 9 tosses is not an integer.

By process of elimination, the answer must be 5 tosses. We can confirm that if she tosses 2 tails in her first 4 tosses, then 50% of her tosses are tails, and if her final toss is a tail, then she has tossed 3 tails out of 5 tosses, which is 60% tails, as required.

18. Four of the angle measurements 62° , 85° , 99° , 108° , 114° are the measures of the angles in the same quadrilateral. Which angle measure is not?

- (A) 62° (B) 85° (C) 99° (D) 108° (E) 114°

Source: 2025 Gauss Grade 7 #18

Primary Topics: Geometry and Measurement

Secondary Topics: Angles | Quadrilaterals

Answer: D

Solution:

The sum of the four angle measurements in a quadrilateral is 360° .

The sum of the five angle measurements given is $62^\circ + 85^\circ + 99^\circ + 108^\circ + 114^\circ = 468^\circ$.

The difference, $468^\circ - 360^\circ = 108^\circ$, is the measure of the angle that is not in the quadrilateral.

(We may confirm that $62^\circ + 85^\circ + 99^\circ + 114^\circ = 360^\circ$.)

19. Ten students each receive a card numbered with a different integer from 10 to 19. The students are each given the checklist shown and they check off each box that describes their number.

- ☐ Odd Number
- ☐ Even Number
- ☐ Prime Number
- ☐ Composite Number
- ☐ Perfect Square

How many students check off exactly two boxes?

- (A) 10 (B) 9 (C) 8 (D) 7 (E) 6

Source: 2025 Gauss Grade 7 #19

Primary Topics: Counting and Probability

Secondary Topics: Logic | Prime Numbers | Counting

Answer: B

Solution:

Each integer is either an odd number or it is an even number.

Thus, each of the 10 students will check off exactly one box from the first pair of boxes.

Similarly, every integer greater than 1 is either a prime number or it is a composite number, and so each of the 10 students will check off exactly one box from the second pair of boxes.

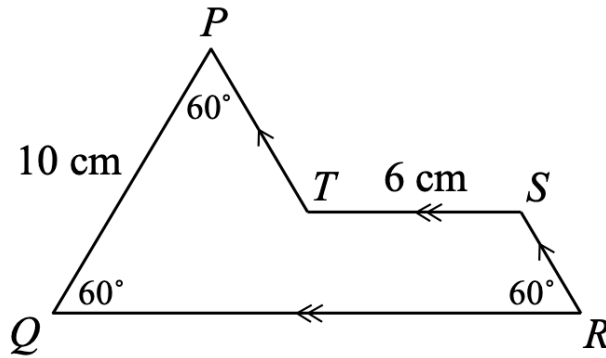
Thus, every student checks off exactly 2 of the first 4 boxes.

This means that every student checks off exactly 2 of the 5 boxes if their card *is not* numbered with a perfect square, and they check off exactly 3 of the 5 boxes if their card *is* numbered with a perfect square.

The card numbered 16 is the only card numbered with a perfect square, and so each of the remaining $10 - 1 = 9$ students have a card that is not numbered with a perfect square.

Thus, there are 9 students that check off exactly two boxes.

20. Figure $PQRST$ is shown below.



In the figure, $\angle PQR = \angle QRS = \angle TPQ = 60^\circ$. Also, PT is parallel to SR and TS is parallel to QR . If $PQ = 10\text{cm}$ and $TS = 6\text{cm}$, the perimeter of figure $PQRST$ is
 (A) 42cm (B) 36cm (C) 40cm (D) 38cm (E) 44cm

Source: 2025 Gauss Grade 7 #20

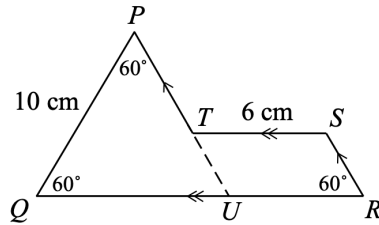
Primary Topics: Geometry and Measurement

Secondary Topics: Angles | Measurement | Perimeter | Polygons

Answer: A

Solution:

We begin by extending PT to point U on QR , as shown.



Since PT is parallel to SR , then TU is parallel to SR .

Similarly, since TS is parallel to QR , then TS is parallel to UR .

Since TU is parallel to SR and TS is parallel to UR , then $TURS$ is a parallelogram, and so $TU = SR$ and $UR = TS = 6\text{ cm}$.

In $\triangle PQU$, the sum of the measures of the three angles is 180° , and so $\angle PUQ = 180^\circ - 60^\circ - 60^\circ = 60^\circ$.

Thus, $\triangle PQU$ is an equilateral triangle, and so $QU = PU = PQ = 10\text{ cm}$.

The perimeter of $PQRST$ is

$$\begin{aligned}
 PQ + QR + SR + TS + PT &= PQ + QU + UR + SR + TS + PT \quad (\text{since } QR = QU + UR) \\
 &= PQ + QU + UR + TU + TS + PT \quad (\text{since } SR = TU) \\
 &= PQ + QU + UR + PT + TU + TS \quad (\text{reordering some sides}) \\
 &= PQ + QU + UR + PU + TS \quad (\text{since } PT + TU = PU) \\
 &= 10\text{ cm} + 10\text{ cm} + 6\text{ cm} + 10\text{ cm} + 6\text{ cm} \\
 &= 42\text{ cm}
 \end{aligned}$$

21. Three circles have radii 1 cm, 5 cm, and x cm. If the mean (average) area of the three circles is $30\pi\text{cm}^2$, the value of x is
- (A) 64 (B) 5 (C) 24 (D) 8 (E) 2

Source: 2025 Gauss Grade 7 #21

Primary Topics: Geometry and Measurement

Secondary Topics: Circles | Measurement | Averages

Answer: D

Solution:

A circle with radius 1 cm has area $\pi \times (1\text{ cm})^2 = \pi\text{ cm}^2$.

A circle with radius 5 cm has area $\pi \times (5\text{ cm})^2 = 25\pi\text{ cm}^2$.

A circle with radius x cm has area $\pi \times (x\text{ cm})^2 = x^2\pi\text{ cm}^2$.

Since the mean area of the three circles is $30\pi\text{ cm}^2$, then the sum of the areas of the three circles is $3 \times 30\pi\text{ cm}^2 = 90\pi\text{ cm}^2$.

Since $\pi + 25\pi + x^2\pi = 26\pi + x^2\pi$, we get $26\pi + x^2\pi = 90\pi$ or $x^2\pi = 64\pi$ or $x^2 = 64$, and so $x = 8$ (since $x > 0$).

22. Each of three doors is painted one colour: either black or white or gold. Each colour is equally likely to be chosen for each door. What is the probability that at least one colour is not used?

- (A) $\frac{7}{9}$ (B) $\frac{5}{9}$ (C) $\frac{20}{27}$ (D) $\frac{2}{9}$ (E) $\frac{2}{3}$

Source: 2025 Gauss Grade 7 #22

Primary Topics: Counting and Probability

Secondary Topics: Probability | Logic

Answer: A

Solution:

Solution 1:

The probability that at least one colour is not used is equal to 1 minus the probability that all three colours are used.

Each of the three doors can be painted one of three different colours, and so there are $3 \times 3 \times 3 = 27$ different ways to paint the doors with no restrictions.

If all 3 colours are used, then the first door can be painted 3 different colours, the second door can be painted 2 different colours, and the third door must be painted the 1 remaining colour.

The probability that all three colours are used is thus $\frac{3 \times 2 \times 1}{27} = \frac{6}{27}$, and so the probability that at least one colour is not used is $1 - \frac{6}{27} = \frac{27}{27} - \frac{6}{27} = \frac{21}{27}$, which is equal to $\frac{7}{9}$.

Solution 2:

The probability that at least one colour is not used is equivalent to the probability that at most two different colours are used.

Thus, the probability that at least one colour is not used is equal to the sum of the probability that exactly one colour is used and the probability that exactly two different colours are used.

We begin by determining the probability that exactly one colour is used.

Each of the three doors can be painted one of three different colours, and so there are $3 \times 3 \times 3 = 27$ different ways to paint the doors with no restrictions.

If exactly one colour is used, there are 3 choices for the colour and each door must be painted with that colour, and so the probability that exactly one colour is used is $\frac{3}{27}$.

If exactly two colours are used, then two doors are painted the same colour and the remaining door is painted a different colour.

There are 3 choices for the first colour and then 2 choices for the second colour, and thus $3 \times 2 = 6$ ways to choose the two colours. Once the two colours are chosen, the doors can be painted with these colours in 3 different ways. For example, if the colour chosen for two of the doors is black (B) and the colour for the remaining door is white (W), the doors can be painted BBW, BWB or WBB.

Therefore, the number of ways to paint the doors with exactly two different colours is $6 \times 3 = 18$,

and so the probability that exactly two different colours are used is $\frac{18}{27}$.

Finally, the probability that at least one colour is not used is $\frac{3}{27} + \frac{18}{27} = \frac{21}{27} = \frac{7}{9}$.

23. Suppose a, b and c are the last three digits of the six-digit integer $N = 111\,abc$. If N is divisible by 18, how many possibilities are there for N ?
- (A) 50 (B) 55 (C) 56 (D) 110 (E) 112

Source: 2025 Gauss Grade 7 #23

Primary Topics: Number Sense

Secondary Topics: Divisibility | Counting | Digits

Answer: C

Solution:

Solution 1:

We are being asked to find the number of multiples of 18 between 111 000 and 111 999 inclusive.

Since $18 \times 6166 = 110\,988$ and $18 \times 6167 = 111\,006$, then the smallest multiple of 18 greater than or equal to 111 000 is 111 006. (We can find the numbers 6166 and 6167 by dividing 111 000 by 18 and rounding the result both down and up to the nearest integer.)

Since $18 \times 6222 = 111\,996$ and $18 \times 6223 = 112\,014$, then the largest multiple of 18 less than or equal to 111 999 is 111 996. (We can similarly find the numbers 6222 and 6223 by dividing 111 999 by 18 and rounding the result both down and up to the nearest integer.)

All multiples of 18 from 18×6167 to 18×6222 inclusive satisfy the given conditions, and so there are $6222 - 6167 + 1 = 56$ different possibilities for N .

Solution 2:

Since $N = 111abc$ is divisible by 18, then N is divisible by both 2 and 9 since 2 and 9 have no factors in common and $2 \times 9 = 18$.

Since N is divisible by 2, then N is even and so its units digit, c , must equal 0, 2, 4, 6, or 8.

An integer is divisible by 9 exactly when the sum of its digits is divisible by 9.

Thus, the sum of the digits of N , which is equal to $1 + 1 + 1 + a + b + c = a + b + c + 3$, must be a multiple of 9.

The smallest possible value of $a + b + c + 3$ is 3 (when $a = b = c = 0$) and its largest possible value is 30 (when $a = b = c = 9$).

The multiples of 9 in this range are 9, 18 and 27, and so $a + b + c = 6$ or $a + b + c = 15$ or $a + b + c = 24$.

To summarize to this point, $N = 111abc$ is divisible by 18 exactly when c is equal to 0, 2, 4, 6, or 8, and $a + b + c = 6$ or $a + b + c = 15$ or $a + b + c = 24$.

We proceed by setting c equal to each of the possible even digits and determining the number of possible values for a and b .

When $c = 0$, we get $a + b = 6$ or $a + b = 15$ or $a + b = 24$.

When $a + b = 6$, the possible values for the digits a and b (written as ordered pairs (a, b)) are $(0, 6)$, $(1, 5)$, $(2, 4)$, $(3, 3)$, $(4, 2)$, $(5, 1)$, and $(6, 0)$. (Alternately, when $a + b = 6$, a can equal any integer from 0 to 6 inclusive and then $b = 6 - a$.)

Thus, there are 7 ordered pairs of digits (a, b) when $c = 0$ and $a + b = 6$, and so there are 7 possible values of N .

When $a + b = 15$, the possible pairs of digits (a, b) are $(6, 9)$, $(7, 8)$, $(8, 7)$, and $(9, 6)$, and so there are 4 possible values of N .

When $a + b = 24$, there are no possible pairs of digits (a, b) since $a + b$ is at most $9 + 9 = 18$.

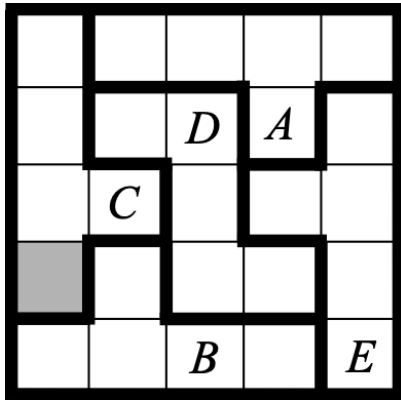
When $c = 2$, we get $a + b = 6 - 2 = 4$ or $a + b = 15 - 2 = 13$ or $a + b = 24 - 2 = 22$.

We continue to count the number of pairs of digits (a, b) given each of the possible values of c and summarize those results in the tables below.

$c = 0$	$a + b = 6$	$a + b = 15$	$a + b = 24$	Number of values of N
Number of pairs (a, b)	7	4	0	11
$c = 2$	$a + b = 4$	$a + b = 13$	$a + b = 22$	Number of values of N
Number of pairs (a, b)	5	6	0	11
$c = 4$	$a + b = 2$	$a + b = 11$	$a + b = 20$	Number of values of N
Number of pairs (a, b)	3	8	0	11
$c = 6$	$a + b = 0$	$a + b = 9$	$a + b = 18$	Number of values of N
Number of pairs (a, b)	1	10	1	12
$c = 8$	$a + b = -2$	$a + b = 7$	$a + b = 16$	Number of values of N
Number of pairs (a, b)	0	8	3	11

Therefore, the number of possibilities for N is $11 + 11 + 11 + 12 + 11 = 56$.

24. In the diagram, each row, each column, and each shape shown by the thick lines must contain the letters *A*, *B*, *C*, *D*, and *E*.



A five by five grid with thick lines dividing the grid into five different shapes. Each shape is made up of five connected squares from the grid. Numbering the rows in the grid 1 through 5 from top to bottom and the columns 1 through 5 from left to right, the squares making up each of the five shapes are as follows:

1. Squares 1 through 4 in column 1 and square 3 in column 2.
2. Squares 2 through 5 in row 1 and square 4 in row 2.
3. Squares 2 and 3 in row 2, square 3 in row 3, and squares 3 and 4 in row 4.
4. Square 2 in row 4 and squares 1 through 4 in row 5.
5. Square 3 in column 4 and squares 2 through 5 in column 5.

There are letters in five of the squares as follows:

- *A* is in row 2, column 4.
- *B* is in row 5, column 3.
- *C* is in row 3, column 2.
- *D* is in row 2, column 3.
- *E* is in row 5, column 5.

The square in row 4, column 1 is shaded.

If each square contains exactly one letter, what letter must be placed in the shaded square?

- (A) *A* (B) *B* (C) *C* (D) *D* (E) *E*

Source: 2025 Gauss Grade 7 #24

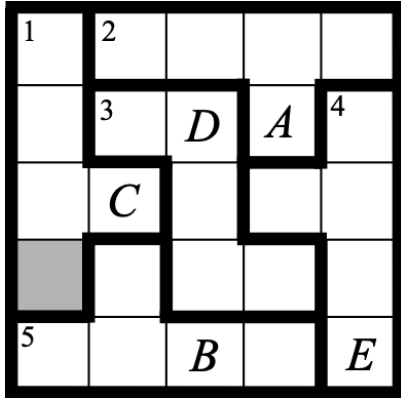
Primary Topics: Other

Secondary Topics: Logic | Counting | Games

Answer: B

Solution:

We begin by naming shapes shown by the thick lines 1, 2, 3, 4, and 5, as shown in the diagram.



Hide/Reveal Description of Shapes 1 - 5

We also adopt the notation [row number, column number] to refer to the contents of specific squares in the diagram.

For example, we are given that [2, 3] is *D* (this is in shape 3), and [2, 4] is *A* (this is in shape 2).

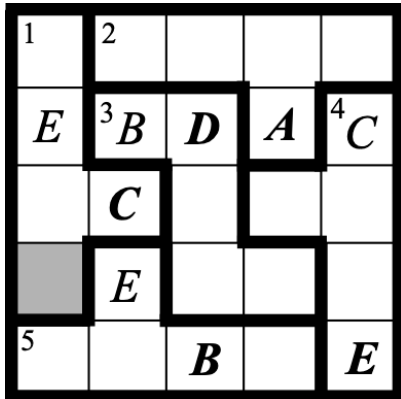
It is important to note that the solution that follows is one of many different ways to arrive at the final answer.

Since [3, 2] is *C*, then [2, 1] cannot be *C* (shape 1 already has *C*), and [2, 2] cannot be *C* (column 2 already has *C*), and thus the *C* in row 2 must be [2, 5].

Since row 5 already has an *E*, then the *E* in shape 5 cannot be in row 5, and thus [4, 2] is *E*.

Row 2 is missing *B* and *E* and since column 2 has an *E*, then [2, 1] is *E* and [2, 2] is *B*.

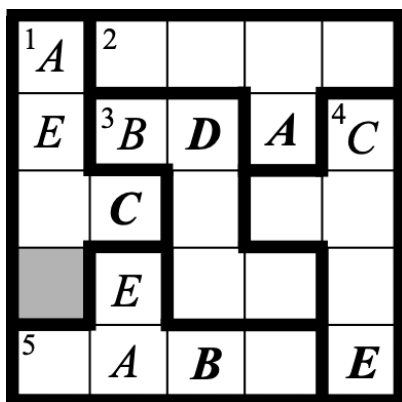
The letters determined to this point are included in the diagram below.



Since shape 2 already contains *A* and row 1 is missing *A*, then [1, 1] is *A*.

Also, column 4 already contains *A* and so [5, 4] is not *A* which means that [5, 2] is *A* (since shape 5 is missing *A*).

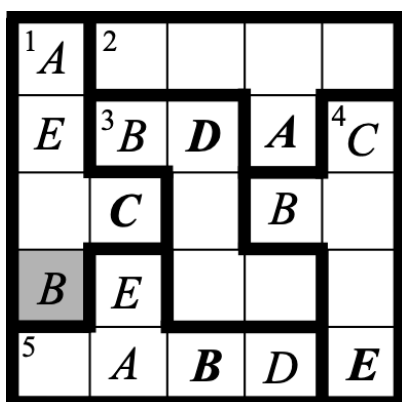
The letters determined to this point are included in the diagram below.



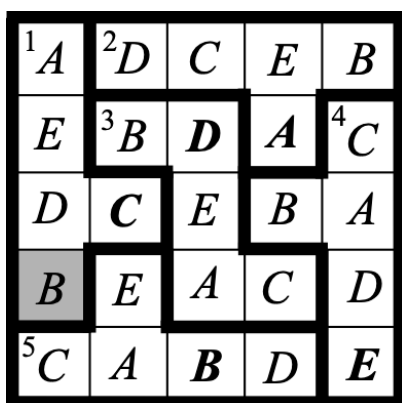
Shape 1 must contain *D* somewhere in column 1 and so $[5, 1]$ is not *D*. Since shape 5 is missing *D*, then $[5, 4]$ is *D*.

Shape 4 is missing *A*, *B* and *D*. Since column 4 already contains *A* and *D*, then $[3, 4]$ cannot be *A* or *D*, and thus $[3, 4]$ is *B*.

Finally, shape 1 is missing *B* and *D*. Since row 3 contains *B*, then $[3, 1]$ cannot be *B*, and so the letter in the shaded square, $[4, 1]$, is *B* as shown.



The completed diagram is included below. Given the initial 5 letters and their locations, there is exactly one way to complete this diagram.



Shape 1: Squares 1 through 4 in column 1 and square 3 in column 2.

Shape 2: Squares 2 through 5 in row 1 and square 4 in row 2.

Shape 3: Squares 2 and 3 in row 2, square 3 in row 3, and squares 3 and 4 in row 4.

Shape 4: Square 2 in row 4 and squares 1 through 4 in row 5.

Shape 5: Square 3 in column 4 and squares 2 through 5 in column 5.

25. In an *arithmetic grid*, adjacent numbers increase by a fixed integer $a > 0$ moving left to right within each row. Also, adjacent numbers increase by a fixed integer $b > 0$ moving top to bottom within each column. For example, the grid shown is a 3×3 arithmetic grid with $a = 2$ and $b = 5$.

1	3	5
6	8	10
11	13	15

Suppose that an 8×8 arithmetic grid has a 1 in the top left corner, and a number less than 75 in the bottom right corner. How many such grids have a 45 somewhere in column 5?

- (A) 6 (B) 3 (C) 7 (D) 5 (E) 4

Source: 2025 Gauss Grade 7 #25

Primary Topics: Algebra and Equations

Secondary Topics: Patterning/Sequences/Series | Expressions | Counting

Answer: A

Solution:

We adopt the notation [row number, column number] to be equal to the contents of a specific square in the grid.

Since $[1, 1]$ is 1 and adjacent numbers increase by a fixed integer $a > 0$ moving left to right within each row, then $[1, 2]$ is $1 + a$, $[1, 3]$ is $1 + 2a$, and so on to the end of the row where $[1, 8]$ is $1 + 7a$.

Similarly, adjacent numbers increase by a fixed integer $b > 0$ moving top to bottom within each column, and so the contents of each square in row 2 is b greater than the adjacent square in row 1.

That is, $[2, 1]$ is $1 + b$, $[2, 2]$ is $1 + a + b$, $[2, 3]$ is $1 + 2a + b$, and so on to the end of the row where $[2, 8]$ is $1 + 7a + b$.

Continuing in this way in row 3, we get $[3, 1]$ is $1 + 2b$, $[3, 2]$ is $1 + a + 2b$, $[3, 3]$ is $1 + 2a + 2b$, and so on to $[3, 8]$ which is equal to $1 + 7a + 2b$.

We continue this pattern in the table below, and include each of the entries in column 5 since the focus of the question is on this column.

1	$1 + a$	$1 + 2a$	$1 + 3a$	$1 + 4a$	$1 + 5a$	$1 + 6a$	$1 + 7a$
$1 + b$	$1 + a + b$	$1 + 2a + b$	$1 + 3a + b$	$1 + 4a + b$	$1 + 5a + b$	$1 + 6a + b$	$1 + 7a + b$
$1 + 2b$	$1 + a + 2b$	$1 + 2a + 2b$	$1 + 3a + 2b$	$1 + 4a + 2b$	$1 + 5a + 2b$	$1 + 6a + 2b$	$1 + 7a + 2b$
$1 + 3b$	$1 + a + 3b$	$\vdots$	$\vdots$	$1 + 4a + 3b$	$\vdots$	$\vdots$	$\vdots$
$1 + 4b$	$1 + a + 4b$			$1 + 4a + 4b$			
$1 + 5b$	$1 + a + 5b$			$1 + 4a + 5b$			
$1 + 6b$	$1 + a + 6b$			$1 + 4a + 6b$			
$1 + 7b$	$1 + a + 7b$			$1 + 4a + 7b$			$1 + 7a + 7b$

We are given that the number in the bottom right corner of the grid is less than 75, and so $1 + 7a + 7b < 75$ or $7a + 7b < 74$.

Dividing each term of this inequality by 7, we get $\frac{7a}{7} + \frac{7b}{7} < \frac{74}{7}$ or $a + b < \frac{74}{7}$ and since a and b are positive integers, then $a + b \leq 10$.

Since $a + b \leq 10$, then by multiplying each term by 4, we get $4a + 4b \leq 40$.

In the table above, $[5, 5]$ is $1 + 4a + 4b$. Since $4a + 4b \leq 40$, then $1 + 4a + 4b \leq 41$ and so $[5, 5]$ cannot equal 45.

Further, since b is a positive integer, then each of the entries in column 5 above row 5 is less than $1 + 4a + 4b$ and thus cannot equal 45.

Thus if 45 appears in column 5 of this grid, then it can only appear in rows 6, 7 and 8.

We begin by noting that a and b are positive integers, and since $a + b \leq 10$, then $a \leq 9$ and $b \leq 9$.

If $[6, 5]$ is 45, then $1 + 4a + 5b = 45$ or $4a + 5b = 44$.

If $a = 1$, then $4 \times 1 + 5b = 44$ or $5b = 40$, and so $b = 8$.

We confirm that when $a = 1$ and $b = 8$, then $a + b \leq 10$ and so the number appearing in the bottom right corner of the grid is less than 75.

Thus, in the arithmetic grid with $a = 1$ and $b = 8$, $[6, 5]$ is 45.

If $a = 2$, then $4 \times 2 + 5b = 44$ or $5b = 36$ or $b = \frac{36}{5}$ (which is not an integer), and so there is no such arithmetic grid in which $[6, 5]$ is 45 when $a = 2$.

We could continue substituting the remaining values of a from 3 to 9 to determine for which values of a , $4a + 5b = 44$ and $a + b \leq 10$ and b is a positive integer.

Alternately, we might notice that $4a$ is a multiple of 4, as is 44.

Thus, if $4a + 5b = 44$ then $5b$ must also be a multiple of 4 and so b must be a multiple of 4.

The only remaining positive integer b for which $b \leq 9$ and b is a multiple of 4 is $b = 4$.

When $b = 4$, we get $4a + 5 \times 4 = 44$ or $4a = 24$ and so $a = 6$.

We again confirm that when $a = 6$ and $b = 4$, then $a + b \leq 10$ and so the number appearing in the bottom right corner of the grid is less than 75.

In the arithmetic grid with $a = 6$ and $b = 4$, $[6, 5]$ is 45.

Thus, there are exactly 2 such grids in which 45 appears in row 6, column 5.

If $[7, 5]$ is 45, then $1 + 4a + 6b = 45$ or $4a + 6b = 44$ or $2a + 3b = 22$.

In this case, we similarly notice that $2a$ is a multiple of 2, as is 22.

Thus, if $2a + 3b = 22$ then $3b$ must also be a multiple of 2 and so b must be a multiple of 2.

We proceed by checking values of b which are equal to positive even integers.

When $b = 2$, we get $2a + 3 \times 2 = 22$ or $2a = 16$ and so $a = 8$.

In this case, $a + b \leq 10$ and so the number appearing in the bottom right corner is less than 75.

In the arithmetic grid with $a = 8$ and $b = 2$, $[7, 5]$ is 45.

When $b = 4$, we get $2a + 3 \times 4 = 22$ or $2a = 10$ and so $a = 5$.

In this case, $a + b \leq 10$.

In the arithmetic grid with $a = 5$ and $b = 4$, $[7, 5]$ is 45.

When $b = 6$, we get $2a + 3 \times 6 = 22$ or $2a = 4$ and so $a = 2$.

In this case, $a + b \leq 10$.

In the arithmetic grid with $a = 2$ and $b = 6$, $[7, 5]$ is 45.

When $b = 8$, we get $2a + 3 \times 8 = 22$ which is not possible since $a > 0$.

Thus, there are exactly 3 such grids in which 45 appears in row 7, column 5.

If $[8, 5]$ is 45, then $1 + 4a + 7b = 45$ or $4a + 7b = 44$.

We notice that $4a$ is a multiple of 4, as is 44.

Thus, if $4a + 7b = 44$ then $7b$ must also be a multiple of 4 and so b must be a multiple of 4.

We proceed by checking the two possible values of b , namely $b = 4$ and $b = 8$.

When $b = 4$, we get $4a + 7 \times 4 = 44$ or $4a = 16$ and so $a = 4$.

In this case, $a + b \leq 10$ and so the number appearing in the bottom right corner is less than 75.

In the arithmetic grid with $a = 4$ and $b = 4$, $[8, 5]$ is 45.

When $b = 8$, we get $4a + 7 \times 8 = 44$ which is not possible since $a > 0$.

Thus, there is exactly 1 such grid in which 45 appears in row 8, column 5, and so there are $2 + 3 + 1 = 6$ grids in total.
